

Data Understanding

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Data Understanding

2.1 Gathering Data

The dataset was obtained from Kaggle. According to the dataset author, the data was scraped from Craigslist in approximately January 2020 and compiled into a single CSV file. The raw file has the following dimensions:

- File size: 558.44 MB
- Total rows: 384,977
- Total columns: 22

2.2 Describing Data

The dataset contains a mix of numeric, categorical, text, and geographic feature types. The target variable for this project is price.

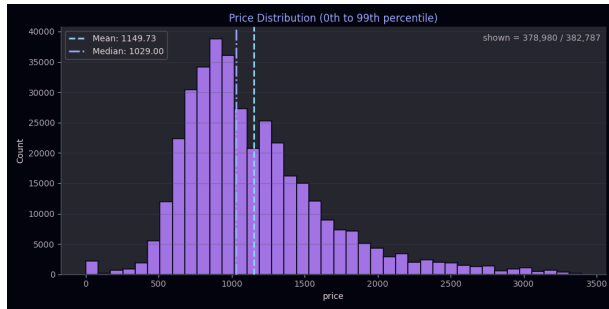
Feature Type	Features	Notes
Numeric	price, sqfeet, beds, baths	Target: price
Categorical (Binary)	cats_allowed, dogs_allowed, smoking_allowed, wheelchair_access, electric_vehicle_charge, comes_furnished	Boolean amenity flags
Categorical (Multi)	region, type, laundry_options, parking_options	Multi-class labels
Geographical	lat, long, state, region	Used for location clustering
Text / URL	id, url, region_url, image_url, description	Excluded from modeling

2.3 Exploring Data

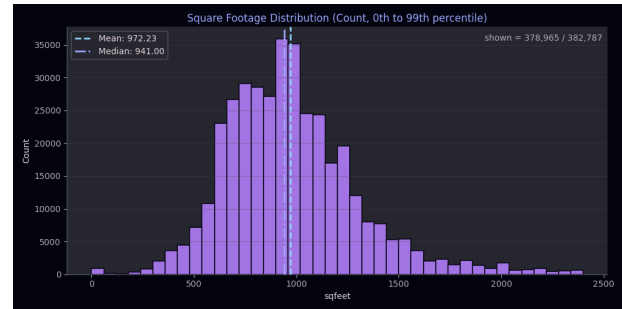
Initial exploratory analysis revealed the following key patterns and distributions within the dataset. Apartments are the most common property type, followed by houses and townhouses. Most properties fall below the third quartile values of 1,150 square feet and \$1,395 in price, and most listings are below two bathrooms and two bedrooms at the third quartile level. The dataset is described as largely pet friendly, while furnished listings, EV charging availability, and wheelchair accessibility are all reported as minority cases. About one quarter of listings are noted as smoke friendly.

Figure 1

Exploratory data analysis histograms for key housing variables.



(a) Price distribution



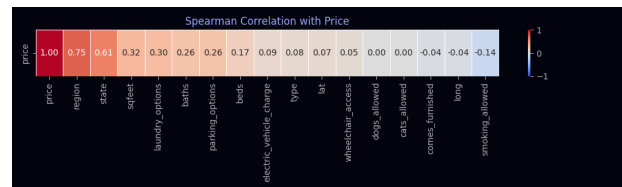
(b) Square footage distribution

Figure 2

Correlation Test - Numeric Features vs Price



(a) Pearson



(b) Spearman

2.4 Verifying Data Quality

A structured quality assessment was conducted to identify missing values, duplicates, pricing anomalies, and structural issues before data preparation began.

2.4.1 Missing Values

The following columns were found to contain missing values:

Feature	Missing Count	Missing %
laundry_options	79,026	20.53%
parking_options	140,687	36.54%
lat	1,918	0.50%
long	1,918	0.50%

2.4.2 Anomalies and Outliers

- Properties listed at \$0 were identified and removed. These may represent listings open to negotiation, but they introduce noise into price modeling.
- Properties with less than 1 sq ft were identified and removed.

- Properties with 0 bedrooms and 0 bathrooms were retained, as these may represent valid land rental listings.
- Price outliers were noted and will be addressed in Section 3.2